Smart Road Accident Hotspot Analysis & Risk Prediction

Comprehensive Technical & Analytical Executive Report | UK STATS19 (2021–2025)

1. Executive Summary

Road traffic collisions represent one of the primary global causes of preventable trauma and economic loss. The Smart Road Accident Hotspot Analysis and Risk Prediction System is an end-to-end data analytics and predictive platform designed to help civil engineers, transport authorities, and emergency services transition from reactive accident response to proactive, targeted hazard prevention.

By harmonizing five continuous years (2021–2025) of official UK Department for Transport (STATS19) records, the system combines unsupervised spatial clustering (DBSCAN in metric British National Grid coordinates) with explainable supervised classification (Balanced Logistic Regression) to identify systemic infrastructure blackspots and isolate the primary physical determinants of severe trauma.

[CORE PRINCIPLE] Road traffic fatalities are not unpredictable accidents; they are systematically patterned events governed by junction geometry, ambient lighting conditions, vehicle vulnerability, and speed physics.

2. Dataset Scale & System Architecture

The operational database and analytics pipeline integrate records across three normalized levels of cardinality:

Domain / Table	Volume (Records)	Scope & Description
Accidents (accident)	100,000	Cleaned, validated UK collisions (2021–2025) with spatial BNG coords.
Vehicles (vehicle)	183,908	Individual vehicles involved (type, driver sex/age band, manoeuvre).
Casualties (casualty)	130,370	Injured victims (demographics,

		casualty class, injury severity).
Spatial Hotspots (hotspot)	705	Validated high-risk clusters identified via DBSCAN spatial clustering.

3. Key Empirical Findings from Visual Analytics

3.1 The Exposure vs. Lethality Paradox

Visual analysis demonstrates a critical divergence between crash frequency and crash lethality:

• Peak Volume Windows: Total collisions peak sharply during weekday rush hours (08:00–09:00 and 16:00–18:00 on Fridays) in dry daylight. These are dominated by minor rear-ends and lane conflicts (>75% slight severity).

• Peak Severity Windows: The proportion of fatal and severe collisions surges between 01:00 AM and 04:00 AM (reaching 33%–35%), driven by elevated vehicular velocity, fatigue, and intoxication on unlit roads.

3.2 The Darkness Multiplier (Environmental Analysis)

While fine weather and daylight hold >80% of aggregate volume due to travel exposure, unlit darkness constitutes the single highest environmental risk factor. Driving in darkness without streetlights multiplies severe collision likelihood to 34.95% (compared to 23.19% in daylight), representing a +50.7% relative risk jump.

3.3 Kinetic Energy & Speed Limit Scaling

Injury severity progression is non-linear and conforms to quadratic kinetic energy scaling ($E_k = 0.5 * m * v^2$):

• 20 mph Urban Zones: 0.47% fatal rate.

• 30 mph Urban Roads: 0.87% fatal rate.

• 60 mph Single Carriageways: 3.84% fatal rate (a 4.4x jump over 30 mph) and 34.49% severe rate.

Single carriageways generate 72.6% of crashes due to lack of central physical dividers and high closure speeds.

3.4 Vulnerable Road User (VRU) Disproportionality

Passenger cars comprise 68% of involved vehicles. However, motorcyclists exhibit a 33.83% severe injury rate upon impact, being nearly 1.5x more likely to suffer fatal or

incapacitating trauma than enclosed vehicle occupants. Pedal cyclists and pedestrians suffer the highest severe trauma proportion at complex urban intersections.

4. Spatial Clustering Methodology (DBSCAN)

Rather than aggregating by artificial administrative lines (boroughs or postal codes) which suffer from Modifiable Areal Unit Problem (MAUP), the platform uses DBSCAN (eps = 150–750m, MinPts = 25) in metric British National Grid space:

1. Follows Natural Geometry: Hugs the actual curvature of highway interchanges, roundabout slipways, and arterial merges.
2. Filters Random Noise: Automatically ignores isolated one-off accidents, focusing solely on recurring structural blackspots.
3. High Concentration: Demonstrates that over 13% of all national severe casualties concentrate in just 705 localized geographic clusters.

5. Severity Prediction Model Formulation

The operational severity predictor employs Balanced Logistic Regression. Although named 'regression' for historical reasons, it is a true classification algorithm mapping feature vectors through the sigmoid link function P(Severe | x) = 1 / (1 + e^-z). Logistic regression was chosen over ensemble black-box algorithms because:

• Clinical Explainability: Weights translate directly into quantifiable odds ratios (e.g. Darkness log-odds +0.518 => 1.68x risk).
• Calibrated Probabilities: Outputs a continuous, well-calibrated 0–100% risk gauge for scenario simulation.
• Sub-Millisecond In-Memory Execution: Serialized into lightweight JSON metadata, allowing zero-dependency execution in Node.js.

6. Strategic Recommendations & Action Plan

Priority Tier	Policy / Engineering Action	Target Location & Expected Impact
Priority 1	Targeted Hotspot Engineering: Anti-skid resurfacing, clear junction sightlines, automated speed/red-light cameras.	Top 705 DBSCAN clusters (multi-lane roundabouts & merges). Directly mitigates 13% + of recurring national trauma.
Priority 2	Single Carriageway Speed & Median Barriers: Center rumble	60 mph unseparated rural single carriageways. Prevents

	strips and speed limit reviews.	head-on impacts, addressing the 4.4x fatality multiplier.
Priority 3	Active Illumination Upgrades: Smart LED street lighting and high-reflectivity road studs.	High-speed corridors with darkness severe rate >30%. Mitigates the +50.7% darkness penalty.
Priority 4	Vulnerable Road User Segregation: Physically separated cycleways and Dutch-style roundabouts.	Dense urban intersections. Shields motorcyclists and cyclists who carry over 80% of hospitalization burden.